Extreme-value (GEV) flood statistics: data and sources

Team 4 | Wallonia + Liège–Maastricht–Aachen triangle | All links opened on 29 Oct 2026 | **External sources, NOT challenge data** (Annex 2 "EFAS" = hazard maps, a different product)

Wording for the brief: GEV estimates *how rare* a flood is (its return period) under an assumed stable climate. It does **not** predict floods. Say "how rare" or "return period", never "prediction".

1. Usable datasets, best first for a one-day build

Gauge / area	Source	Years, time step	Access
Vesdre – Chaudfontaine (6228) Ourthe – Tabreux (5921) Amblève – Martinrive (6621) Meuse – Lixhe (5436)	SPW Wallonia hydrometry, KiWIS web service hydrometrie.wallonie.be	Annual max from 1967 / 1969 / 1974 / 1996 (60 / 58 / 53 / 31 yrs). Hourly, daily and hydrological-year max also available.	JSON, no login. **Tested from Python.** Licence not verified.
Meuse – Borgharen Dorp, Sint Pieter Noord (NL)	Rijkswaterstaat DDL, Python package `rws-ddlpy` deltares.github.io/ddlpy	Daily means from at least 1980 (start not verified); sub-daily for 2021.	No key. **Tested.**
Inde – Eschweiler, Rur, Wurm (DE)	LANUK NRW checked discharge series opengeodata.nrw.de/.../hydro/q/	1950s–2020s, ~10–15 min.	CSV, open licence (DL-DE Zero 2.0). **Tested.** Wurm (Herzogenrath 2): July 2021 peak missing.
Rainfall return levels, **Jalhay**	RMI / KMI IDF table opendata.meteo.be/.../IDF_table_INS63038_fr.pdf	10 min – 30 days, 2–200 yr, with 95% CI.	PDF, open.
July 2021 rain gauges + RADFLOOD21 radar	RMI case page opendata.meteo.be/cases/202107.php	13–16 July 2021	Open (Zenodo; licence not checked).
Modelled discharge, whole area	EFAS historical (CEMS) ewds.climate.copernicus.eu/datasets/efas-historical	1992 – now, 6-hourly, ~1.5 km	**Needs EWDS account.**
Gridded rainfall	E-OBS v33.0e surfobs.climate.copernicus.eu	1950–2025, daily, 0.1°	NetCDF, non-commercial use only.
German design rainfall	DWD KOSTRA-DWD-2020 opendata.dwd.de/.../KOSTRA/	Not checked	Open directory.

Not usable in one day / not verified: GRDC (data by request), EStreams (11.5 GB, CC BY-NC, Belgian coverage not verified), Geul at Meerssen (no open download found), WVER and KNMI portals.

2. Published numbers to compare against

- **ENW, "Hoogwater 2021 – Feiten en Duiding", §2.4 pp. 21–22** – enwinfo.nl (PDF)
  Borgharen, annual maxima (GEV, Gumbel, GRADE): **~1:100 to 1:200 yr**. Summer half-year GEV: **~1:600** (CI ~1:100 to 1:10,000). Sint Pieter peak 3,260 m^3/s (3,200–3,400), p. 30. Geul, Geleenbeek, Roer: 1:100 to 1:1,000.
- **Rijkswaterstaat, "Analyse hoogwater Maas juli 2021"** – open.rijkswaterstaat.nl
  Borgharen Dorp peak **~3,284 m^3/s**, about **1:125 yr** (official working curve).
- **Journée et al. 2023, HESS 27:3169 (RMI)** – hess.copernicus.org/articles/27/3169/2023
  1–3-day totals well above the 200-yr level, **up to +90%** for 2–3-day totals in the Vesdre basin. Up to 3 h: ≤ 1:30 yr. So: **long soaking rain, not a cloudburst**. RMI 200-yr levels for Jalhay: 108.8 mm (1 d), 138.6 mm (2 d), 150.6 mm (3 d).
- **Tradowsky et al. 2023, Climatic Change 176:90** (peer-reviewed World Weather Attribution study) – doi.org/10.1007/s10584-023-03502-7
  Regionally ≥ 1 in 400 yr; climate change made the rain **1.2–9× more likely** and **3–19% heavier**. No local Meuse return periods.

- **Rongen et al. 2024, HESS 28:2831** – hess.copernicus.org/articles/28/2831/2024
  Borgharen: 120 yr (annual max) and 600 yr (summer). These "were derived including the July 2021 event itself"; before the event it would have looked rarer.
- **Ludwig et al. 2023, NHESS 23:1287** (Ahr, Germany: method example only) – nhess.copernicus.org/articles/23/1287/2023
  Adding historical floods back to 1804 changes the return period a lot (Altenahr ~HQ10,000).
- **Wallonia, "Annexe F – Ajustement GEV" (Oct 2021)** – wallonie.be (PDF)
  14 pages of GEV plots per gauge (hourly discharge, 95% CI, 2021 peak). Plots only, no tables; parent report not named.

Not verified: Dewals et al. 2021 (Water), not opened; Chaudfontaine Q100 ≈ 255 m^3/s (symposium slide only); Deltares/RWS "Evaluation of discharge extremes in the Meuse" (Dec 2023), results not read.

3. Traps to state in the brief

1. **Chaudfontaine 2021 is an estimate.** SPW stores 676.5 m^3/s; the gauge went off its scale, and ENW gives ~520 m^3/s (model estimate).
2. **Daily means hide the peak.** Borgharen: 3,017 (daily mean) vs 3,284 m^3/s (peak). Prefer hourly/instant maxima.
3. **Record length.** With 50–60 years, anything beyond ~1:200 is very uncertain (RMI stops at 200 yr). Always show a confidence range.
4. **Fit with and without 2021** and show both.
5. **Block choice matters** (calendar year, hydrological year from 1 Nov, summer half-year). Say which one you used.
6. **scipy sign:** `scipy.stats.genextreme` uses $c = -\xi$. A heavy tail ($\xi > 0$) appears as $c < 0$.

4. Minimal plan (one day)

1. **Gauges:** Vesdre at Chaudfontaine (strongest story), Ourthe at Tabreux (2021: 542 m^3/s vs 433 in 1993), Meuse at Borgharen (cross-border, has published numbers). Optional: Inde at Eschweiler (329.5 m^3/s in July 2021).
2. **Fetch:** one KiWIS request per Walloon gauge (annual or hydrological-year max); `ddlpy` for Borgharen.
3. **Fit:** `genextreme.fit` without 2021, then with 2021; return period of the 2021 peak $T = 1 / (1 - F(x))$; 1,000 bootstrap resamples for the CI.
4. **Check:** compare with ENW and Rijkswaterstaat (Borgharen) and Annexe F (Wallonia). Log the comparison in the AI sheet.

Sources found with AI help (Claude) and each one opened and checked on 29 Oct 2026. Researchers: re-open and cite every source with URL and access date before using it in the brief (Annex 3).